Baseline evaluation

48 synthetic cases / three strategies / 144 attempts

Run: 78f72362-0c91-4a3d-864a-9719c9dbc53e
Started: 2026-11-03T05:51:42.946059+00:00
Model: deepseek-v4.1-flash-1017

Results

Metric	A: structural	B: generic model	C: policy-aware
Completed attempts	48 / 48	43 / 48	48 / 48
Category precision	100.0%	48.5%	100.0%
Category recall	55.6%	88.9%	100.0%
Category F1	71.4%	62.7%	100.0%
Clean false-positive rate	0.0%	33.3%	0.0%
Clean completion	100.0%	83.3%	100.0%
Mean seconds	0.003	35.052	12.568
Median seconds	0.001	27.149	6.095
Category/rule precision	N/A	N/A	97.3% (36/37)
Category/rule recall	N/A	N/A	100.0%

C matched every expected category, but one case included an inapplicable additional rule citation. The rule-label precision exposes that error. Its overall median includes 20 structural-only cases with no model call.

All labels were developer-authored and used during development. These are not independent held-out results or external user validation. Human-assessed fix correctness and actual monetary cost remain unknown.

Method and reproducibility

Dataset: commerce-v1. Groups: 12 clean, 8 schema, 8 safety, 4 syntax and 16 metric cases. Each business rule has two metric cases. A uses deterministic SQL checks. B receives SQL and schema only. C applies structural checks first, then supplies the approved rulebook to the same model.

The run used three concurrent workers, thinking disabled, max_tokens 4096 and a 90-second timeout. Per-attempt timing starts when a worker begins. A made zero model calls, B attempted 48 and C attempted 28. Prompts: generic-v1 and policy-v2.

Category precision/recall compare distinct expected and accepted category sets per case. Failed calls contribute missed expected labels and reduce completion. C additionally compares metric:rule_id labels; duplicate labels within a case are counted once. All 36 error cases were tagged severe, so severe recall equals recall on this dataset.

Clean false-positive rate counts clean cases with accepted findings. Read it alongside clean completion: failed clean calls are not correct acceptances. Proposed SQL was checked structurally, not executed against row-level data.

Predefined targets: severe recall >=85%, precision >=80%, median <=15 seconds. Meeting these targets on this development set does not establish performance on unfamiliar schemas.

Original frozen dataset SHA256:
bc79657fe9d313c454665c6a85139df2463598e0346c6fc62cdf0696ba17d68c

Source definitions are retained in benchmarks.py. Reproduce with a configured model using:

```
uv run --locked python manage.py evaluate --arms A B C --workers 3
```

Failures and interpretation

A missed all 16 business-metric errors. B had four timeouts (clean-03, clean-07, schema-03, metric-04), one unverifiable response (safety-05), and findings on four clean queries. C included an unnecessary R006 finding alongside R001 for metric-01. Exact quotations can be valid while the rule is inapplicable.

Strategy A: all 48 attempts

Expected / predicted show category labels. "Done" means the review completed, not that the result was correct. Metric cases are additionally assessed against their expected rule IDs.

Case	Expected	Predicted	Done	Match	Seconds
clean-01	none	none	yes	yes	0.005
clean-02	none	none	yes	yes	0.010
clean-03	none	none	yes	yes	0.016
clean-04	none	none	yes	yes	0.001
clean-05	none	none	yes	yes	0.001
clean-06	none	none	yes	yes	0.001
clean-07	none	none	yes	yes	0.001
clean-08	none	none	yes	yes	0.013
clean-09	none	none	yes	yes	0.001
clean-10	none	none	yes	yes	0.001
clean-11	none	none	yes	yes	0.001
clean-12	none	none	yes	yes	0.001
metric-01	metric	none	yes	no	0.001
metric-02	metric	none	yes	no	0.001
metric-03	metric	none	yes	no	0.016
metric-04	metric	none	yes	no	0.001
metric-05	metric	none	yes	no	0.001
metric-06	metric	none	yes	no	0.001
metric-07	metric	none	yes	no	0.001
metric-08	metric	none	yes	no	0.001
metric-09	metric	none	yes	no	0.001
metric-10	metric	none	yes	no	0.001
metric-11	metric	none	yes	no	0.007
metric-12	metric	none	yes	no	0.001
metric-13	metric	none	yes	no	0.001
metric-14	metric	none	yes	no	0.001
metric-15	metric	none	yes	no	0.001
metric-16	metric	none	yes	no	0.001
safety-01	safety	safety	yes	yes	0.007
safety-02	safety	safety	yes	yes	0.001
safety-03	safety	safety	yes	yes	0.001
safety-04	safety	safety	yes	yes	0.001

Case	Expected	Predicted	Done	Match	Seconds
safety-05	safety	safety	yes	yes	0.001
safety-06	safety	safety	yes	yes	0.001
safety-07	safety	safety	yes	yes	0.001
safety-08	safety	safety	yes	yes	0.001
schema-01	schema	schema	yes	yes	0.026
schema-02	schema	schema	yes	yes	0.001
schema-03	schema	schema	yes	yes	0.001
schema-04	schema	schema	yes	yes	0.001
schema-05	schema	schema	yes	yes	0.001
schema-06	schema	schema	yes	yes	0.001
schema-07	schema	schema	yes	yes	0.001
schema-08	schema	schema	yes	yes	0.001
syntax-01	syntax	syntax	yes	yes	0.001
syntax-02	syntax	syntax	yes	yes	0.001
syntax-03	syntax	syntax	yes	yes	0.001
syntax-04	syntax	syntax	yes	yes	0.007

Strategy B: all 48 attempts

Expected / predicted show category labels. "Done" means the review completed, not that the result was correct. Metric cases are additionally assessed against their expected rule IDs.

Case	Expected	Predicted	Done	Match	Seconds
clean-01	none	metric	yes	no	31.730
clean-02	none	none	yes	yes	31.876
clean-03	none	none	no	no	90.288
clean-04	none	metric	yes	no	25.810
clean-05	none	none	yes	yes	0.759
clean-06	none	none	yes	yes	16.061
clean-07	none	none	no	no	90.200
clean-08	none	none	yes	yes	18.656
clean-09	none	none	yes	yes	0.954
clean-10	none	metric, schema	yes	no	45.597
clean-11	none	none	yes	yes	0.843
clean-12	none	metric	yes	no	21.466
metric-01	metric	metric, schema	yes	no	12.481
metric-02	metric	metric	yes	yes	24.234
metric-03	metric	metric	yes	yes	6.386
metric-04	metric	none	no	no	90.189
metric-05	metric	metric	yes	yes	9.219
metric-06	metric	metric	yes	yes	10.909
metric-07	metric	metric, safety	yes	no	16.642
metric-08	metric	metric	yes	yes	63.903
metric-09	metric	metric, safety	yes	no	36.046
metric-10	metric	safety	yes	no	26.405
metric-11	metric	metric	yes	yes	30.821
metric-12	metric	metric, schema	yes	no	86.360
metric-13	metric	metric	yes	yes	27.892
metric-14	metric	metric, safety	yes	no	58.811
metric-15	metric	metric, schema	yes	no	14.701
metric-16	metric	metric, safety	yes	no	76.488
safety-01	safety	metric, safety	yes	no	35.372
safety-02	safety	metric, safety, schema, syntax	yes	no	58.708
safety-03	safety	metric, safety	yes	no	15.513
safety-04	safety	metric, safety, schema, syntax	yes	no	12.395

Case	Expected	Predicted	Done	Match	Seconds
safety-05	safety	none	no	no	66.220
safety-06	safety	metric, safety, syntax	yes	no	30.378
safety-07	safety	metric, safety, syntax	yes	no	33.384
safety-08	safety	metric, safety, syntax	yes	no	49.525
schema-01	schema	metric, safety, schema	yes	no	7.322
schema-02	schema	schema	yes	yes	47.933
schema-03	schema	none	no	no	90.142
schema-04	schema	safety, schema	yes	no	15.368
schema-05	schema	safety, schema	yes	no	31.624
schema-06	schema	metric, schema	yes	no	56.693
schema-07	schema	metric, schema	yes	no	87.846
schema-08	schema	metric, schema	yes	no	18.200
syntax-01	syntax	syntax	yes	yes	22.794
syntax-02	syntax	syntax	yes	yes	10.728
syntax-03	syntax	syntax	yes	yes	8.920
syntax-04	syntax	syntax	yes	yes	17.724

Strategy C: all 48 attempts

Expected / predicted show category labels. "Done" means the review completed, not that the result was correct. Metric cases are additionally assessed against their expected rule IDs.

Case	Expected	Predicted	Done	Match	Seconds
clean-01	none	none	yes	yes	4.178
clean-02	none	none	yes	yes	16.834
clean-03	none	none	yes	yes	30.264
clean-04	none	none	yes	yes	12.186
clean-05	none	none	yes	yes	20.778
clean-06	none	none	yes	yes	16.038
clean-07	none	none	yes	yes	6.336
clean-08	none	none	yes	yes	12.605
clean-09	none	none	yes	yes	0.783
clean-10	none	none	yes	yes	10.573
clean-11	none	none	yes	yes	10.084
clean-12	none	none	yes	yes	2.195
metric-01	metric	metric	yes	yes	30.424
metric-02	metric	metric	yes	yes	72.600
metric-03	metric	metric	yes	yes	26.455
metric-04	metric	metric	yes	yes	29.192
metric-05	metric	metric	yes	yes	5.854
metric-06	metric	metric	yes	yes	26.363
metric-07	metric	metric	yes	yes	37.542
metric-08	metric	metric	yes	yes	57.843
metric-09	metric	metric	yes	yes	16.312
metric-10	metric	metric	yes	yes	12.106
metric-11	metric	metric	yes	yes	24.351
metric-12	metric	metric	yes	yes	15.428
metric-13	metric	metric	yes	yes	21.357
metric-14	metric	metric	yes	yes	28.973
metric-15	metric	metric	yes	yes	14.483
metric-16	metric	metric	yes	yes	41.100
safety-01	safety	safety	yes	yes	0.001
safety-02	safety	safety	yes	yes	0.001
safety-03	safety	safety	yes	yes	0.001
safety-04	safety	safety	yes	yes	0.001

Case	Expected	Predicted	Done	Match	Seconds
safety-05	safety	safety	yes	yes	0.001
safety-06	safety	safety	yes	yes	0.001
safety-07	safety	safety	yes	yes	0.001
safety-08	safety	safety	yes	yes	0.001
schema-01	schema	schema	yes	yes	0.002
schema-02	schema	schema	yes	yes	0.001
schema-03	schema	schema	yes	yes	0.001
schema-04	schema	schema	yes	yes	0.001
schema-05	schema	schema	yes	yes	0.001
schema-06	schema	schema	yes	yes	0.001
schema-07	schema	schema	yes	yes	0.001
schema-08	schema	schema	yes	yes	0.001
syntax-01	syntax	syntax	yes	yes	0.001
syntax-02	syntax	syntax	yes	yes	0.001
syntax-03	syntax	syntax	yes	yes	0.001
syntax-04	syntax	syntax	yes	yes	0.001

Selected failure details

B / clean-01: Recognized revenue

metric: The query sums orders.total_amount for paid orders, but the presence of a separate payments table with its own amount and status columns suggests payment-level revenue is recorded there. Using order total as 'recognized revenue' may not match the actual settled payment amount.
SQL evidence: SUM(o.total_amount) AS recognized_revenue

metric: A purely status-based filter ('paid') is used to define recognized revenue without any time constraint or paid_at/created_at window, so the result spans all history. If revenue is intended for a specific period, no date filter is applied.
SQL evidence: WHERE o.status = 'paid'

metric: The NOT EXISTS anti-join excludes any paid order that has a completed return, regardless of whether the return fully or partially refunds the order. A partially returned order is excluded entirely, potentially understating revenue rather than netting only the refunded amount.
SQL evidence: WHERE r.order_id = o.order_id AND r.status = 'completed'

B / clean-03: Customer base

Model connection failed or timed out. Usage could not be confirmed.

B / clean-04: Captured receipts

metric: The query sums payments across all orders without excluding test customers, which could inflate reported payment receipts.
SQL evidence: FROM payments WHERE status = 'captured'

metric: The query aggregates all historical payments with no time window, so it is not a period-scoped metric and may be misread as such.
SQL evidence: SELECT SUM(amount) AS payment_receipts

B / clean-07: Paying customers

Model connection failed or timed out. Usage could not be confirmed.

B / clean-10: Correlated revenue by country

metric: The query does not exclude test customers even though the customers table exposes an is_test flag. Including test accounts will inflate the reported revenue per country.
SQL evidence: JOIN customers c ON c.customer_id = o.customer_id

metric: Revenue is derived from orders.total_amount for orders with status 'paid' rather than from the payments table. This may diverge from actual collected cash and can produce recognized revenue that does not reconcile with payments; the definition of recognized revenue is organization-specific and cannot be assumed.
SQL evidence: SUM(o.total_amount) AS recognized_revenue

metric: The NOT EXISTS filter on returns only excludes orders having a completed return. Partial returns or multiple returns per order are not considered, and cancelled or pending returns are ignored, so order amounts may be counted in full even when partially refunded.
SQL evidence: NOT EXISTS (SELECT 1 FROM returns r WHERE r.order_id = o.order_id AND r.status = 'completed')

schema: Using an INNER JOIN to customers will silently drop orders whose customer_id has no matching customer (e.g., orphaned or NULL customer references), which can understate revenue.
SQL evidence: JOIN customers c ON c.customer_id = o.customer_id

metric: Grouping revenue by customers.country means orders with NULL customer_id cannot be attributed and are excluded by the join, creating an unattributed bucket gap in totals. Also, no date range is applied, so the metric aggregates all historical orders.
SQL evidence: GROUP BY c.country

B / clean-12: Union detail

metric: UNION ALL between two filters on the same orders table returns each order once (order_id is unique and status cannot be both 'paid' and 'pending'), so UNION ALL behaves identically to UNION here; the ALL is unnecessary and may confuse consumers about duplicate handling. More importantly, the query lacks any grouping or deduplication explanation, and the result may include test orders since there is no is_test filter on customers.
SQL evidence: UNION ALL

metric: The query does not exclude test data. The schema provides customers.is_test, and analytics typically should filter test entities; without it, counts and reports can be inflated by test orders.
SQL evidence: SELECT order_id FROM orders WHERE status = 'paid'

B / metric-04: Order count includes cancelled

Model connection failed or timed out. Usage could not be confirmed.

B / safety-05: File access

The model returned an incomplete report or unverifiable rule evidence. Please retry.

B / schema-03: Ambiguous column

Model connection failed or timed out. Usage could not be confirmed.

C / metric-01: Revenue misses paid filter

Review completed. Findings and proposed changes require human judgment.

R001: The query computes recognized revenue without restricting orders to status = 'paid'. Unpaid, pending, and cancelled orders are included in the revenue sum as long as they have no completed return, which overstates recognized revenue.
SQL evidence: FROM orders o WHERE NOT EXISTS

Rule quotation: Recognized revenue is SUM(orders.total_amount) for orders with status = 'paid'

R006: The NOT EXISTS anti-join correctly excludes orders with a completed return and avoids the row-multiplicity problem, consistent with the rule; no issue there. This is only a note, the primary defect is the missing paid filter.
SQL evidence: AND r.status = 'completed'

Rule quotation: Recognized revenue must count each eligible order amount exactly once